

UNIVERSITY OF MORATUWA

Faculty of Engineering



EE4410 – Electrical Services for Buildings

ASSIGNMENT 02 DIALUX SIMULATION

Date of Submission:

08/03/2026

A. T. S. Amasith

210038K

Department of Electrical Engineering

Table of Contents

01.	Introduction.....	3
02.	Modeling and Design Parameters	3
02.1.	Modeling of the Laboratory	3
02.2.	Modeling Assumptions	6
03.	Existing Lighting Arrangement	7
03.1.	Luminaire Specifications	7
03.2.	Spatial Configuration and Layout.....	8
03.3.	Simulation Results	8
03.4.	Identified Issues	10
04.	Proposed Lighting Arrangement.....	11
04.1.	Design Objectives & Strategy.....	11
04.2.	Selection of Luminaire.....	11
04.3.	Spatial Configuration and Layout.....	12
04.4.	Simulation Results	13
05.	Comparative Evaluation.....	15
06.	Conclusion	16

01.Introduction

This report presents a comprehensive lighting redesign for the Electrical Engineering Computer Laboratory, located on the ground floor of the Sumanadasa Building at the University of Moratuwa. The facility serves as a dual-purpose space, supporting both formal laboratory sessions and independent undergraduate study. Consequently, the primary objective of this redesign is to establish a high-performance luminous environment that ensures visual comfort, minimizes fatigue, and enhances productivity in accordance with modern lighting standards.

The project methodology involved the development of an accurate 3D model in DIALux Evo based on precise on-site measurements of the laboratory's geometry and furniture layout. An initial simulation was conducted to evaluate the technical performance of the existing lighting arrangement and identify specific deficiencies. Based on these findings, an improved, energy-efficient lighting system is proposed. This report provides a comparative analysis of the existing and proposed designs, supported by quantitative performance indicators and engineering justifications.

02.Modeling and Design Parameters

02.1. Modeling of the Laboratory

The initial phase of the modeling process involved a detailed site survey to capture the physical parameters of the computer laboratory. Accurate 3D modeling in DIALux Evo requires precise geometric data to ensure that the Inverse Square Law and Cosine Law calculations for light distribution are applied correctly within the virtual environment. The dimensions recorded represent the internal clear distances of the space, accounting for structural elements such as pillars and recessed areas.



Fig 1 – Modeled 3D design of the lab

The exact dimensions of the laboratory are as follows.

Dimension	Length
Length	16.942m (55'7")
Width	9.449m (31'0")
Hight	3.658m (12'0")
Wall Thickness	0.2m (0'0.78")

To ensure the simulation reflects the actual luminous environment, architectural features such as windows, doors, and structural beams were meticulously integrated into the 3D model.



Fig 2 – Front view of the laboratory

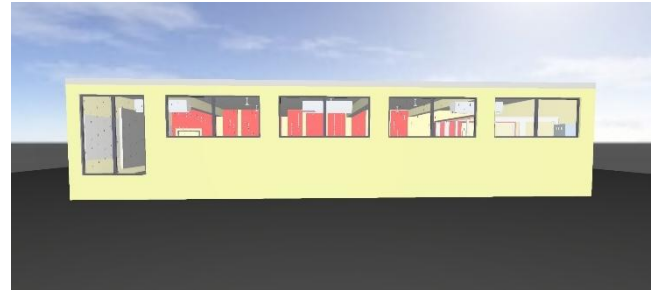


Fig 3 – Back view of the laboratory

The laboratory is designed to accommodate 70 students across primary workstations, with a dedicated administrative zone for technical officers. The interior environment includes a smart screen for instructional purposes and a dual-cooling system comprising air conditioning and backup fans. These elements were integrated into the model.

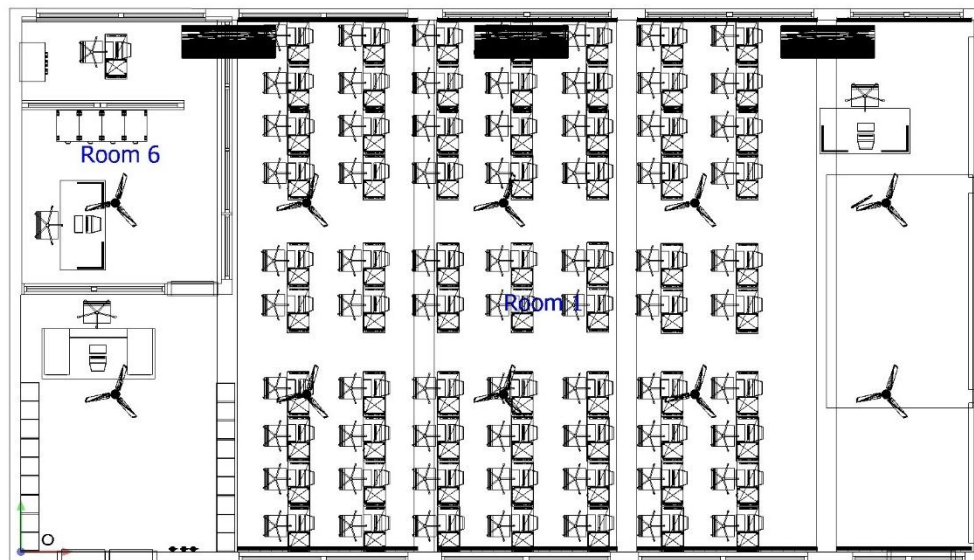


Fig 4 – Top view of the designed model – 2D



Fig 5 – Top view of the designed model – 3D

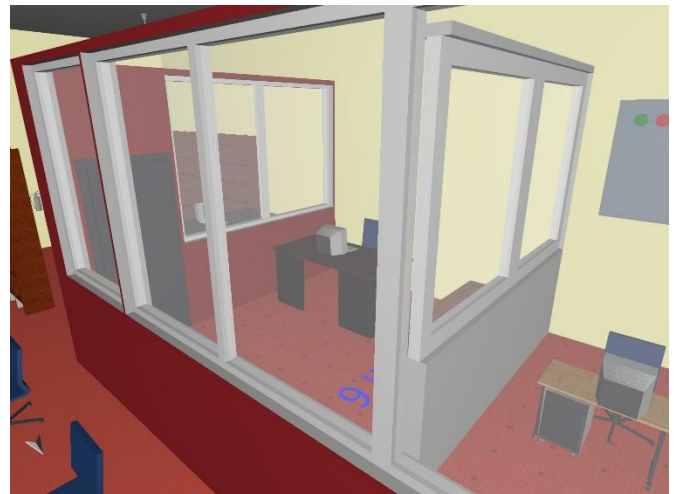


Fig 6 – Designed computer lab inside

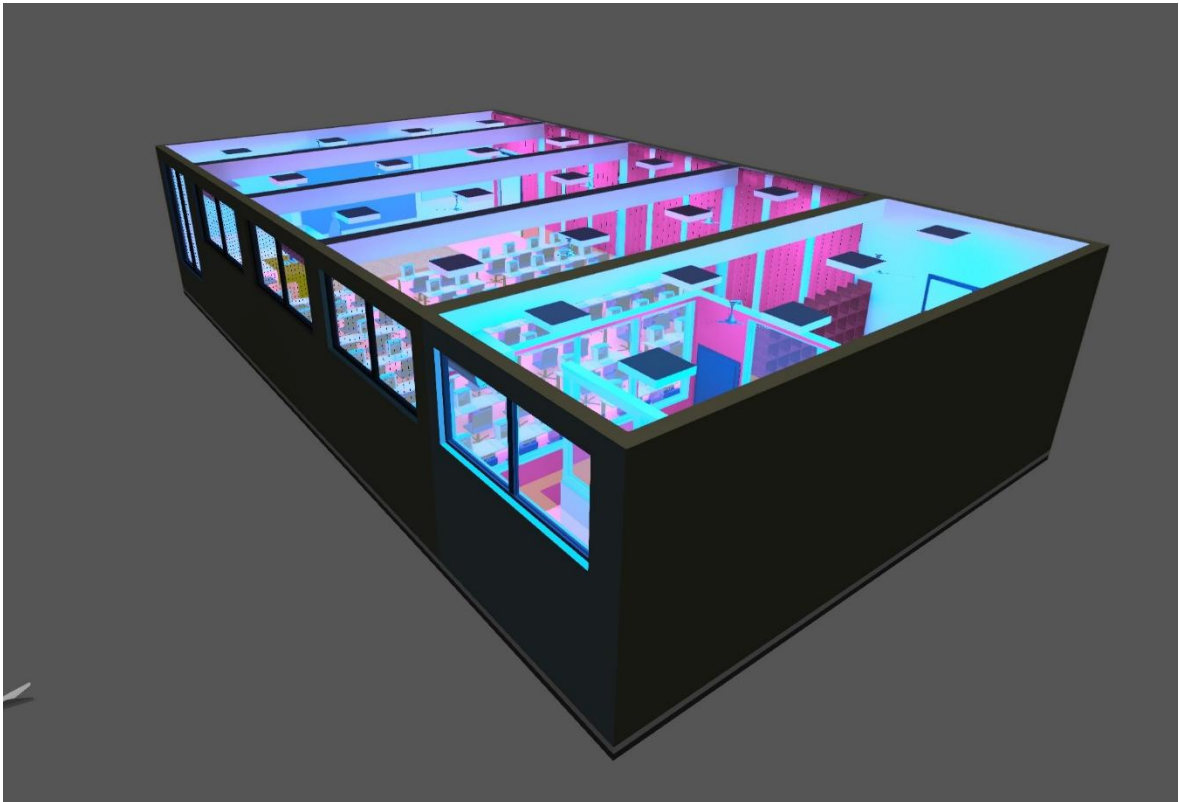
02.2. Modeling Assumptions

Following assumptions were made during the modeling of the computer laboratory.

- Maintenance Factor – A fixed maintenance factor of 0.8 was assumed.
- Reflection Factors – Following values were assumed as the reflection factors of different surfaces.

Type of Surface	Reflection Factor
Walls (Color code - #FEFBC4)	85%
Floor	30%
Table Surfaces	43%
Glass	10%
Computers	44%

- Target Illuminance Level – Assumed to be as 300lux.



03.Existing Lighting Arrangement

To establish a baseline for performance evaluation, the existing lighting infrastructure of the computer laboratory was documented and simulated. This section reflects the current luminaire types, their spatial distribution, and their mounting configuration. By simulating this environment, the technical gaps between the current performance and the required standards can be quantified.

03.1. Luminaire Specifications

The existing lighting installation in the computer laboratory primarily utilizes traditional linear fluorescent technology. The fixtures are recessed into the ceiling grid, featuring aluminum louvers designed to provide basic glare control for computer-based tasks.

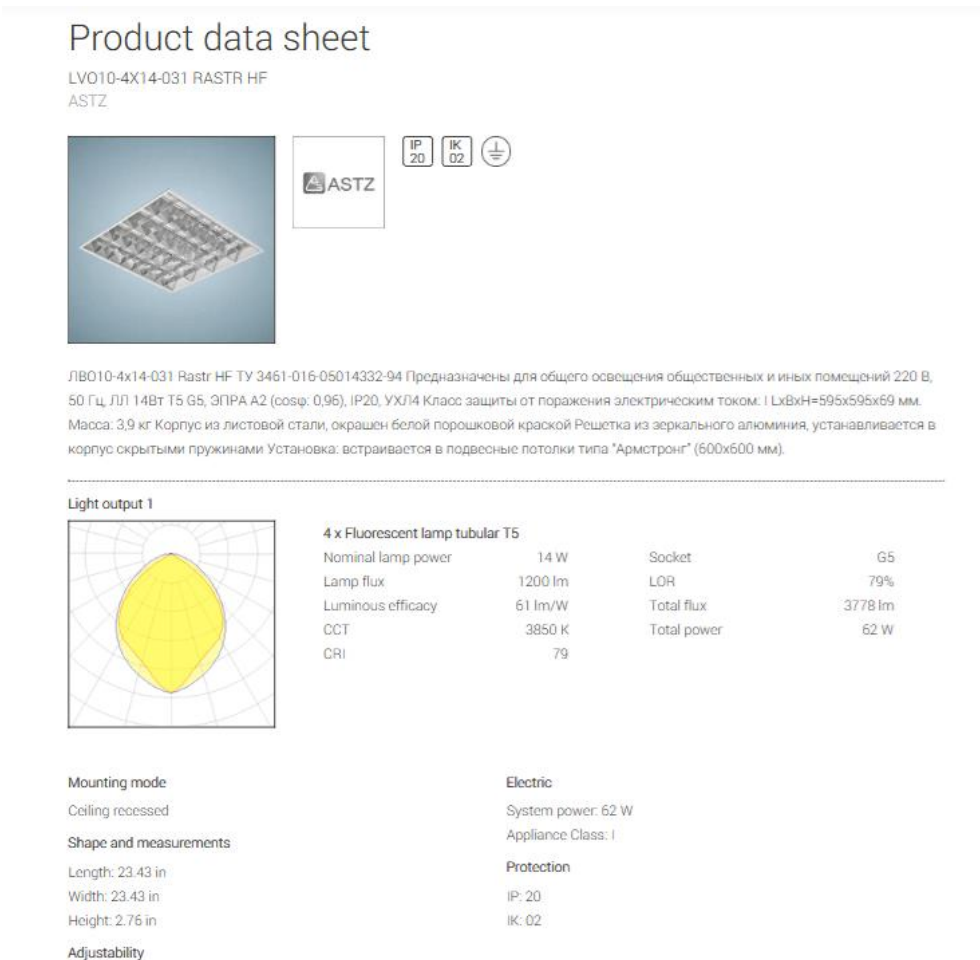


Fig 7 – Datasheet of the existing lamps

03.2. Spatial Configuration and Layout

The existing lighting system is organized in a symmetrical 3×5 grid across the slab of the computer laboratory. This arrangement was chosen in the original design to provide general ambient illumination throughout the 70-student capacity area. Lights were mounted 3.727m above the ground.

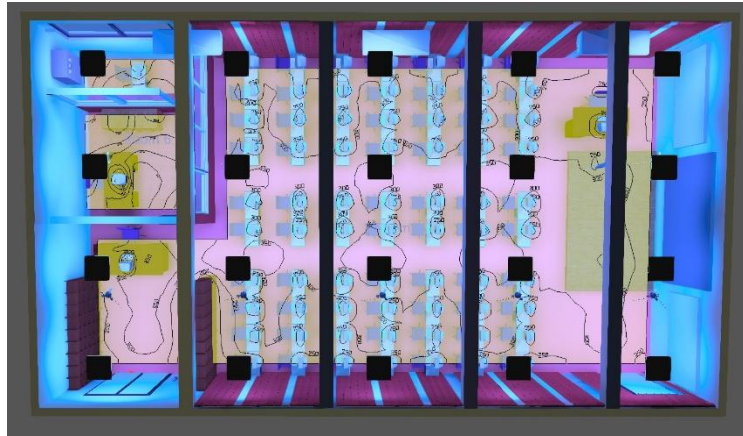


Fig 8 – Existing Light Arrangement

03.3. Simulation Results

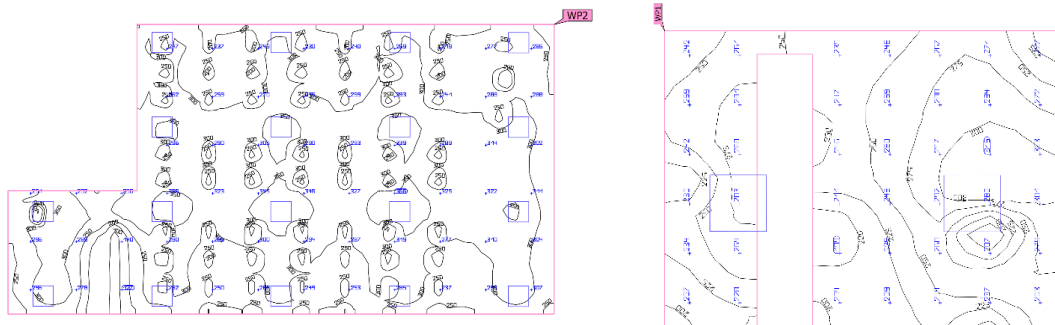
According to the results obtained for the current lighting system it is shown that the required illuminance level of 300lx was not fulfilled.

Building 1 · Storey 1 (Light scene 1)

Calculation objects

Working planes

Properties	E (Target)	E _{min}	E _{max}	U _o (g ₁) (Target)	g ₂	Index
Working plane Perpendicular illuminance (adaptive) Height: 0.800 m, Wall zone: 0.209 m	254 lx (≥ 300 lx) ✗	187 lx	320 lx	0.74 (≥ 0.60) ✓	0.58	WP1
Working plane (Room 1) Perpendicular illuminance (adaptive) Height: 0.800 m, Wall zone: 0.500 m	294 lx (≥ 300 lx) ✗	52.3 lx	382 lx	0.18 (≥ 0.60) ✗	0.14	WP2



Value Chart



Fig 10 – Value Chart - Existing Light Arrangement

Isolux Curves



Fig 11 - Isolux Curves - Existing Light Arrangement

False Colors

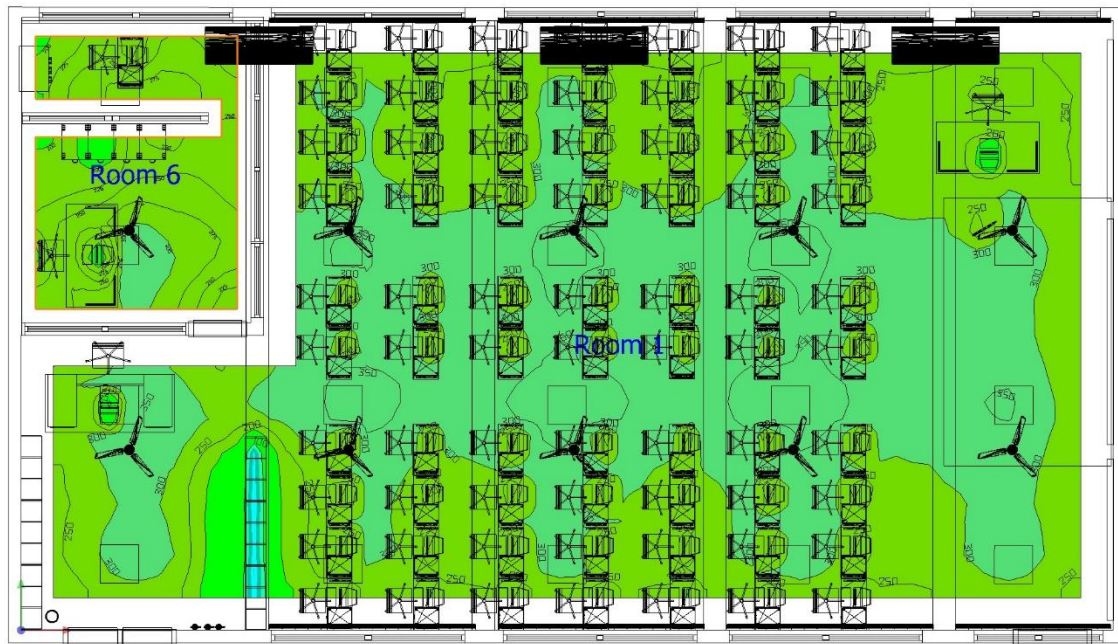


Fig 12 – False Colors - Existing Light Arrangement

03.4. Identified Issues

The simulation reveals that the main laboratory area (Room 1) achieves an average perpendicular illuminance of 294 lx on the working plane. This falls below the target threshold of 300 lx (and significantly below the 500lx recommended by international standards for detailed computer work). This deficit indicates that the current system provides insufficient light for prolonged academic tasks, potentially leading to visual fatigue for the 70 students the lab accommodates.

The most critical technical deficiency identified is the overall uniformity of 0.18 in the main area. The calculated value is vastly lower than the minimum requirement of 0.60. This low uniformity indicates extreme "pooling" of light. With a minimum illuminance of only 52.3 lx compared to a maximum of 382 lx, the laboratory suffers from intense bright spots directly under luminaires and deep shadows in between. For undergraduates performing study work, these high-contrast zones require the pupils to constantly dilate and contract as they look around the room, which is a primary cause of eye strain.

The Unified Glare Rating (UGR) was calculated at 20, exceeding the maximum comfort limit of 19 for educational environments. This suggests that the high-output T5 lamps and the reflective mirror-louvers of the ASTZ Rastr fixtures are likely creating direct glare or "veiling reflections" on computer monitors and the smart screen.

The current system utilizes ASTZ LVO10-4x14-031 fluorescent luminaires with a total power consumption of 1116 W for the main room. The system's luminous efficacy is only 54.8 lm/W, which is poor compared to modern LED technology that typically exceeds 120 lm/W. The Lighting

Power Density (LPD) is 9.58 W/m² on the working plane. While it meets the broad energy estimation limit (1484 kWh/a vs 5000 kWh/a limit), the low efficacy confirms that much of the energy is being wasted compared to the light levels actually delivered.

In summary, the existing lighting arrangement is technically inadequate for a modern computer laboratory. It fails to provide the necessary lux levels for task performance and exhibits unacceptable uniformity, creating a fragmented luminous environment that hinders student productivity and comfort.

04. Proposed Lighting Arrangement

04.1. Design Objectives & Strategy

The proposed lighting design aims to rectify the technical deficiencies identified in the current laboratory setup, specifically addressing the illuminance deficit and the critically low uniformity. The redesign strategy focuses on transitioning from legacy fluorescent technology to high-efficiency LED luminaires to create a high-performance study environment for the 70 students and technical staff.

The primary objectives of the redesigned system are as follows:

- Establish a target average illuminance of at least 300 lx, in accordance with CIBSE and SLS standards for computer laboratory tasks.
- Reduce the Unified Glare Rating to below 19 to ensure visual comfort for users of computer monitors and the smart screen.
- Optimize energy efficiency by significantly reducing the Lighting Power Density while increasing the luminous efficacy of the installation.

04.2. Selection of Luminaire

To meet these objectives, the **Philips RC048B Smart Bright Plus LED Panel** was selected. This luminaire was chosen over the existing ASTZ fluorescent fixtures due to its superior technical profile.

Philips - RC048B+ LED41S PSU W60L60



P	34.0 W
Φ _{Lamp}	4100 lm
Φ _{Luminaire}	4096 lm
η	99.90 %
Luminous efficacy	120.5 lm/W
CCT	3000 K
CRI	100

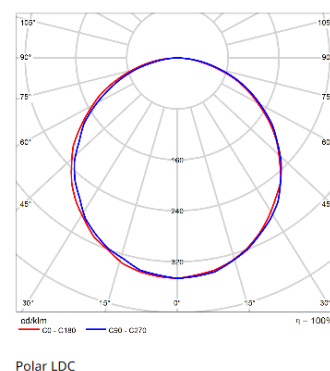


Fig 13 – Luminaire Specifications – Philips RC048B Smart Bright Plus

Engineering Justification for Selection

- **Energy Efficiency:** The selected LED panel provides a luminous efficacy of 120.5 lm/W, which is more than double the efficacy of the existing ASTZ T5 system (54.8 lm/W).
- **Light Output:** With a luminaire flux of 4096 lm, this fixture offers significantly higher output per unit compared to the current 3396 lm fixtures, facilitating the achievement of the 300 lx target for the laboratory.
- **Optical Precision:** The 99.90% efficiency and the uniform Polar Luminous Intensity Distribution (LDC) ensure that almost all generated light is effectively directed toward the work plane, minimizing wasted lumens.
- **Color Quality:** The CRI of 100 ensures perfect color fidelity, which is critical for technical engineering tasks such as identifying color-coded components and wiring in the lab.

04.3. Spatial Configuration and Layout

The proposed lighting system utilizes a total of 22 Philips RC048B LED panels distributed across the entire laboratory floor to ensure comprehensive coverage. By implementing a systematic field arrangement, the design transitions from the fragmented layout of the existing system to a more cohesive spatial configuration that accounts for the laboratory's non-standard geometry. The luminaires are surface-mounted at a uniform height of 3.658 m, optimized to provide sufficient clearance for structural elements and ceiling fans while maximizing the effective beam spread of the integrated PMMA diffusers.

The arrangement strategy focuses on aligning the longitudinal axes of the LED panels with the primary workstation clusters to minimize direct glare and specular reflections on monitor surfaces. By carefully calculating the spacing-to-height ratio, the layout provides a significant boost in average illuminance to 346 lx in the main area and 461 lx in the secondary zone, ensuring the facility meets the 300 lx standard required for detailed technical tasks. This unified approach effectively bridges the luminous gaps found in the current setup, creating a high-performance, energy-efficient environment that supports the academic needs of the 70 students utilizing the space.

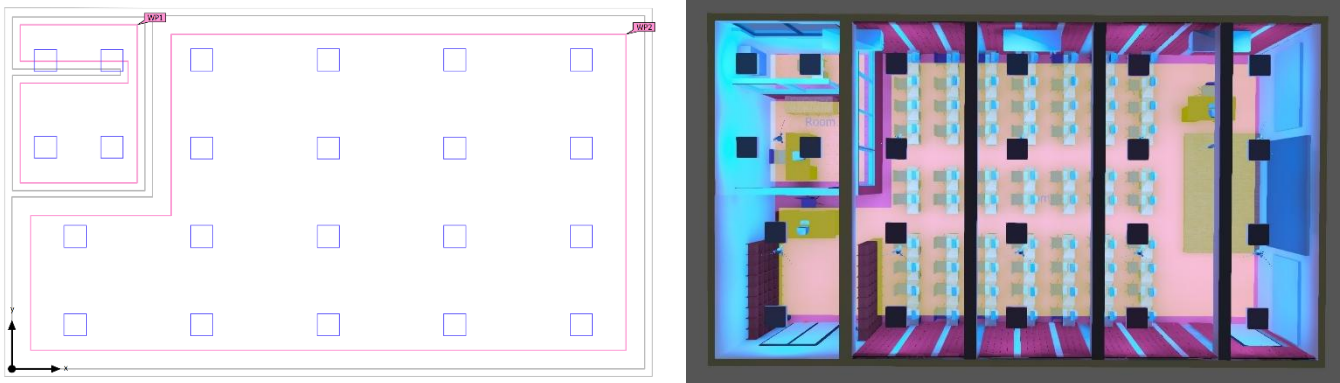


Fig 14 – Proposed Luminaire Layout

04.4. Simulation Results

The redesigned system demonstrates a significant improvement in general illuminance levels compared to the existing installation. The key performance indicators for the laboratory zones are summarized below.

Building 1 · Storey 1 (Light scene 1)

Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_0 (g_1)$ (Target)	g_2	Index
Working plane Perpendicular illuminance (adaptive) Height: 0.800 m, Wall zone: 0.209 m	461 lx (≥ 300 lx) ✓	336 lx	552 lx	0.73 (≥ 0.60) ✓	0.61	WP1
Working plane (Room 1) Perpendicular illuminance (adaptive) Height: 0.800 m, Wall zone: 0.500 m	346 lx (≥ 300 lx) ✓	69.6 lx	449 lx	0.20 (≥ 0.60) ✗	0.16	WP2

Value Chart

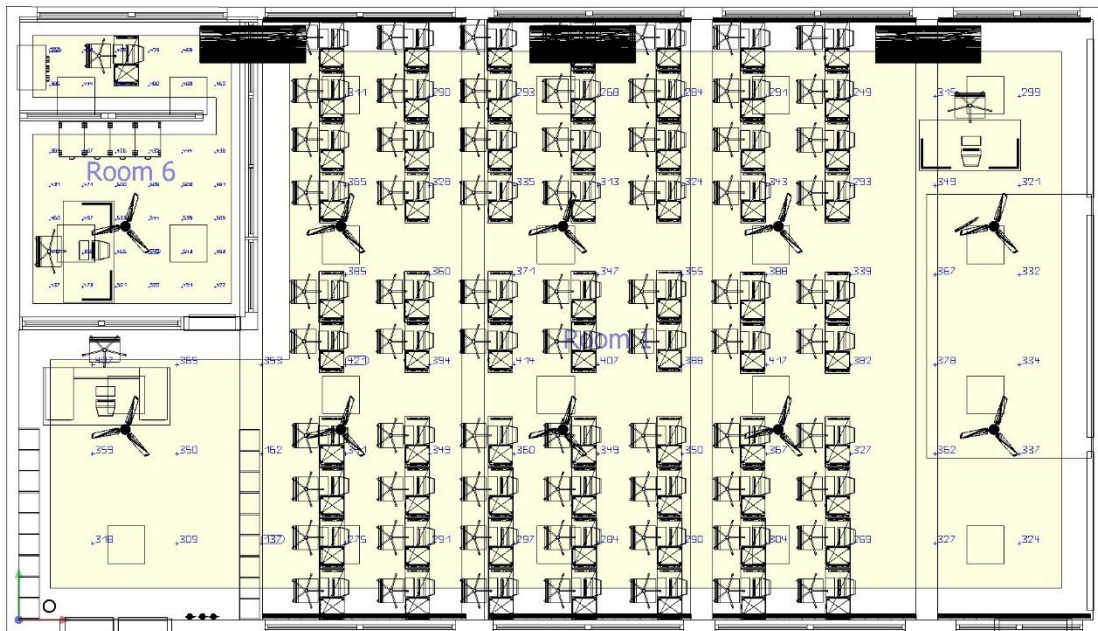


Fig 15 – Value Chart - Proposed Light Arrangement

Isolux Curves



Fig 16 - Isolux Curves - Proposed Light Arrangement

False Colors

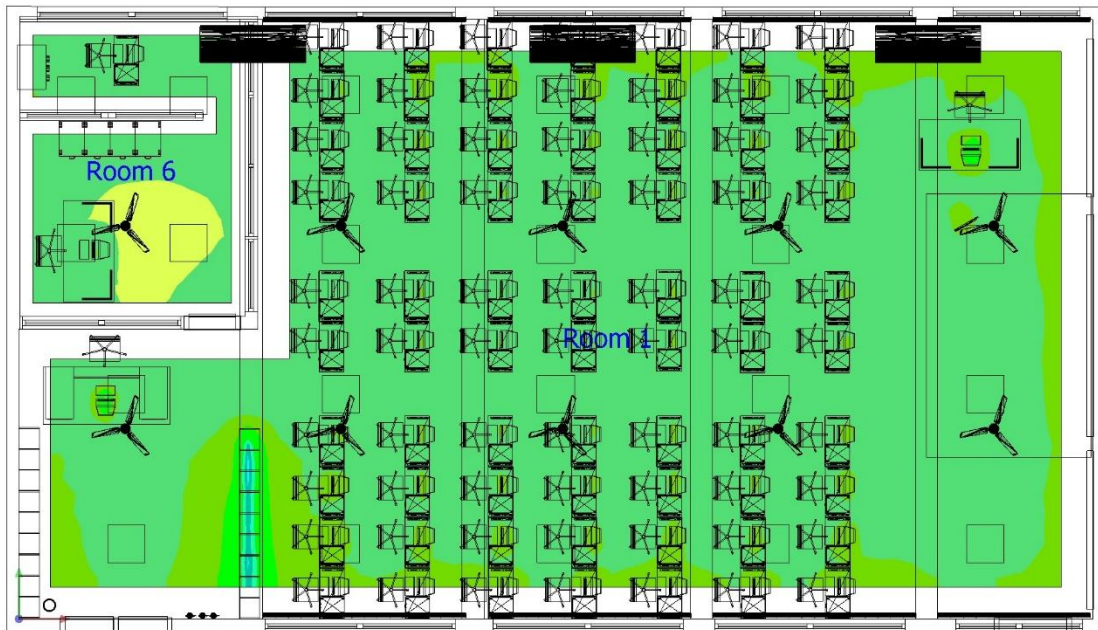


Fig 17 – False Colors - Proposed Light Arrangement

While the proposed design successfully elevates the average perpendicular illuminance to 346 lx, surpassing the 300 lx target, the calculated uniformity of 0.20 in room 1 remains below the recommended 0.60 threshold for educational environments. This deviation is primarily attributed to the following engineering and architectural constraints:

- **Luminance Contrast and Point-Source Effect:** The high output Philips LED panels (4096 lm each) create intense localized concentrations of light directly beneath the fixtures, reaching a maximum of 449 lx. Because the current layout uses only 18 units to cover a large $141.83m^2$ area, the "overlap" between beam angles is insufficient at the 0.800 m work plane, leading to a minimum illuminance of only 69.6 lx in the "inter-luminaire" gaps.
- **Obstructions and Shading:** The presence of ceiling fans, bag racks and air conditioning units in the Sumanadasa building laboratory creates physical barriers that disrupt the ideal distribution of light. These elements cause localized shading, which contributes to the low minimum lux levels recorded in the simulation.
- **Wall Zone and Perimeter Effects:** A 0.500 m wall zone was applied to the calculation, which includes several perimeter workstations. The light fall-off at the edges of the room is more pronounced because the surface-mounted arrangement is restricted by the concrete slab, preventing the placement of fixtures closer to the walls to fill the perimeter shadows.

Therefore, although uniformity does not strictly meet the 0.60 standard, the proposed arrangement is a significant technical improvement over the existing legacy system. The priority was placed on achieving the required 346 lx average to ensure task visibility while working within the mounting limitations of the laboratory's concrete slab roof.

05.Comparative Evaluation

The following table contrasts the As-Is conditions with the proposed redesign across the entire storey, including the primary laboratory and the secondary technician area:

Parameter	Existing Arrangement	Proposed Arrangement	Performance Shift
Total Luminous Flux	67920lm	90112lm	+32.7% Increase
Total Power Consumption	1240W	748W	-39.7% Savings
System Luminous Efficacy	54.8lm/W	120.5lm/W	119.9% Efficiency
Avg. Illuminance (Main Area)	294lx	346lx	17.7% Improvement
Avg. Illuminance (Technician Area)	254lm	461lx	81.5% Improvement

The most significant engineering achievement of this redesign is the doubling of the system's luminous efficacy from 54.8 lm/W to 120.5 lm/W. While the existing system consumes 1,240 W to produce approximately 68,000 lumens, the proposed LED system produces over 90,000 lumens while drawing only 748 W. This reduction in load directly translates to lower operational costs and a smaller carbon footprint for the University of Moratuwa's Sumanadasa building.

The proposed design successfully rectifies the illuminance deficit in both the main laboratory and the technician area bringing both zones well above the 300 lx target. Furthermore, the leap in Color Rendering Index (CRI) from 79 to 100 ensures that technical tasks, such as component identification on PCBs, are performed with maximum visual accuracy.

In the primary laboratory area, the uniformity saw a slight increase from 0.18 to 0.20, though it remains below the 0.60 target due to the specific architectural constraints and the wide spacing of the 18 high-output panels required to keep the total power low. Conversely, the technician area maintains an excellent uniformity of 0.73, proving that in smaller, more concentrated zones, the Philips RC048B panels provide a near-perfectly balanced study atmosphere.

06. Conclusion

The lighting redesign of the computer laboratory successfully transitions the facility from an inefficient, legacy fluorescent system to a high-performance LED infrastructure. By utilizing Philips RC048B+ LED panels, the proposed design achieves an average perpendicular illuminance of 346 lx in the main laboratory and 461 lx in the secondary area, both of which now surpass the minimum 300 lx target for educational computer practice rooms.

One of the most significant engineering outcomes of this project is the 45.2% reduction in energy consumption within the main laboratory, achieved by dropping the total power load from 1116.0 W to 612.0 W. This drastic improvement is a direct result of the superior 120.5 lm/W efficacy and CRI of 100 offered by the new LED panels, which provide a more natural and energy-efficient study atmosphere compared to the existing 54.8 lm/W system.

While the architectural constraints of the Sumanadasa building and the surface-mounting requirements on the concrete slab led to a uniformity of 0.20 in the main hall, the redesign significantly improved the minimum light levels and overall visual comfort for the 70 students using the space. In conclusion, the proposed lighting arrangement provides a technically sound, energy-efficient solution that satisfies the functional requirements of the department while significantly reducing the university's operational energy footprint.